time in increments of 100 milliseconds or 100 seconds, utilising three common cathode display (LTS 543)—DIS-1 through DIS-3.

When selecting a dual stopwatch that is commonly available in sports equipment stores or with online retailers, look for features such as precision, durability, and ease of use. The presented dual stopwatch provides precision to count up to 100 milliseconds and 100 seconds, making it suitable for specific applications. The author's prototype on breadboard is shown in Fig. 1.

Circuit and working

The circuit diagram of the dual stopwatch is shown in Fig. 2. It is built around two NE555 timers (IC1, IC2), three 4033 7-segment display drivers (IC3 through IC5), three common cathode 7-segment displays (DIS-1 through DIS-3), and several other components.

The 4033 IC facilitates easy interfacing with 7-segment displays and is versatile for applications like decade counting, seven-segment decimal displays, watches, and timers.

Timers IC1 and IC2 operate in an astable multivibrator mode, generating millisecond and second pulses, respectively. The outputs of these timers are routed to the three 4033 ICs (IC3 through IC5) to display units, tens, and hundreds digits.

PARTS LIST

Semiconductors:	
IC1, IC2	- NE555 timer
IC3-IC5	- 4033 decade counter
DIS-1-DIS-3	- LTS543 common cathode 7-segement display
LED1	- 5mm red LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
R2	- 33-kilo-ohm
R3	- 56-kilo-ohm
R4	- 5.8-kilo-ohm
R5	- 4.7-kilo-ohm
R6	- 10-kilo-ohm
R7-R9	- 470-ohm
Capacitors:	
C1-C4	- 0.01µF ceramic disk
C2	- 10µF, 16V electrolytic
C3	- 0.1µF ceramic disk
Miscellaneous:	
CON1	- DC socket
S1, S2, S4	- Push-to-on switch
S3	- SPDT switch
	- 5V DC adaptor

Switches S1 and S2 initiate and stop the stopwatch for milliseconds and seconds, Switch S3 enables mode change, and switch S4 resets the stopwatch. A 5V DC socket powers the entire circuit, with LED1 indicating power presence.

In this stopwatch circuit, the delay in seconds is generated using IC2, and the duration of the delay is based on the value of the resistor used. Calculation formula for generating frequency for the astable multivibrator (built around IC2) is:

$F = 1/T = 1.44/(R2 + 2R3)\ C2$

Here R2 is 33k, R3 is 56k, and C2 is 10µF.

Similarly, astable multivibrator IC1 generates millisecond delays. The